

LiSA: Light-Space Atlases for Relightable Gaussian Splatting

Supplementary Material

This supplement gives implementation details (Sec. S1), the evaluation protocol and per-scene results (Sec. S2), additional analyses (Sec. S3) and qualitative results (Sec. S4), and the protocols of the experiments that are still to be run (Sec. S5). Section, table and equation numbers without the prefix “S” refer to the main paper.

S1. Implementation details

Light camera. For every light we build a look-at camera from the light position $\mathbf{p}$ to the scene center $\mathbf{c}$. Its half field of view is the 99.9th percentile, over all Gaussians in front of the light, of the angular extent of their 3σ footprints, enlarged by 5% and capped at 75° ; distant floaters may therefore fall outside the atlas. Depths are measured along the optical axis, relative to $\|\mathbf{c} - \mathbf{p}\|$, and divided by the scene radius R , which is estimated from the training cameras. The texel size τ_0 is the width of one atlas texel at the depth of the scene center, in units of R , and $\tau_k = 2^k \tau_0$.

Pyramid and gather. Each level applies a separable $[1, 4, 6, 4, 1]/16$ filter with zero padding and 2×2 average pooling. Receivers sample every level bilinearly with zero padding, so points outside the light frustum see zero coverage and are treated as lit. In the descriptor, t_k is divided by four and clamped to $[-2, 2]$, and δ_k is multiplied by four and clamped to $[-4, 4]$; $M_{0,k}$ is clamped to $[0, 1]$ and the mean depth uses $\max(M_{0,k}, 10^{-4})$ in the denominator.

Networks. All networks are MLPs with SiLU activations. The flux encoder MLP ϕ has two hidden layers of width 32 and its output bias is initialized to $\text{softplus}^{-1}(1)$. The visibility residual g has two hidden layers of width 64 with a zero-initialized output layer. The transfer network h has two hidden layers of width 128 and outputs $K \cdot 3 \cdot C = 168$ values; its output bias is -7 , so that $L_{\text{tr}} \approx 0$ at the start. The material network has four hidden layers of width 128 and outputs nine RGB responses (one base and eight basis responses); its inputs are the code $\mathbf{f}$, the normal, four-band Fourier encodings of $\mathbf{l}, \mathbf{v}, \mathbf{h}$ and the reflected view direction, five cosines ($\mathbf{n} \cdot \mathbf{l}, \mathbf{n} \cdot \mathbf{v}, \mathbf{n} \cdot \mathbf{h}, \mathbf{v} \cdot \mathbf{h}, \mathbf{l} \cdot \mathbf{v}$) and five highlight hints with $\kappa \in \{8, 32, 128, 512, 2048\}$. The eight light-independent coefficients come from a two-layer MLP of width 64 on $\mathbf{f}$ and an eight-band positional encoding, with a zero-initialized output; the lobe weights of $s(\mathbf{x})$ come from a network of the same shape with output bias -6 . The normal channels of $\mathbf{f}$ are detached wherever $\mathbf{f}$ enters a network as a latent code, so the learned normals receive gradients only through the directional terms in which they appear (cosines,

the reflected view direction and the $\mathbf{n} \cdot \mathbf{l}$ input of the flux encoder).

Optimization. Density uses the screen-space gradient rule of 3DGS (threshold $2 \cdot 10^{-4}$) from iteration 500 to 25k, with opacity resets every 3k iterations and a cap of 400k Gaussians. During the first 2k iterations, pixels outside the ground-truth mask supervise only coverage. The light pass is first used after the warm-up, i.e., at iteration 8001. In stage III, the networks start from their stage-II weights with fresh optimizer state; geometry, cameras and the light normalization are frozen.

Real captures. For each training camera we optimize a rotation about its calibrated optical center, starting at iteration 2k with a learning rate decayed from 10^{-3} to 10^{-5} . With fixed centers and free rotations, a global translation of the scene can be absorbed by re-aiming all cameras. After every camera update we project the rotation corrections off this three-dimensional gauge in closed form and translate the scene by the corresponding offset, which leaves the image position of the object center unchanged in every training view and keeps the reconstruction in the calibrated frame. Test cameras and lights always use the official calibration.

S2. Evaluation protocol and per-scene results

Images are evaluated at 512 px on the longest side (256 px for SSS-GS). The GS³ scenes are linear HDR renderings displayed with $\gamma = 2.2$ on a white background; the other benchmarks are displayed as provided on black. SSIM uses an 11×11 Gaussian window ($\sigma = 1.5$) with zero padding, and LPIPS uses the VGG network with inputs in $[-1, 1]$. Each method’s renderings are compared with one common set of 8-bit targets, namely the test images as encoded by our evaluation pipeline. The targets produced by the baselines’ own pipelines differ from these by at most one intensity level, except on three real captures (CatSmall, CupFabric, Pikachu), where 6–10% of the channel values of GS³, SSD-GS and RNG differ by up to 42 levels. Re-scoring against the common targets changed per-scene PSNR by at most 0.04 dB and scene-averaged PSNR by at most 0.005 dB, and it changed no ranking. Table S1 lists all per-scene results.

S3. Additional analyses

Transfer share. Table S2 reports, for every scene, the share of linear radiance that the final model routes through the transfer term. It is lowest on the opaque synthetic objects (5.6% on Drums to 9.3% on Lego), higher on fur

Table S1. **Per-scene results** (PSNR $\uparrow$ / SSIM $\uparrow$ / LPIPS $\downarrow$), common targets, original test calibration, seed 0. Best PSNR per scene in **bold**.

Benchmark	Scene	RNG	OLAT-GS	GS ³	SSD-GS	LiSA (ours)
NRHints (real)	Cat	20.99 / 0.773 / 0.287	22.43 / 0.768 / 0.206	19.12 / 0.717 / 0.232	18.09 / 0.705 / 0.247	25.33 / 0.821 / 0.171
	CatSmall	18.74 / 0.792 / 0.212	31.58 / 0.949 / 0.081	33.28 / 0.967 / 0.076	33.83 / 0.967 / 0.079	33.02 / 0.961 / 0.082
	CupFabric	21.41 / 0.847 / 0.201	36.40 / 0.979 / 0.049	35.16 / 0.976 / 0.053	35.98 / 0.979 / 0.054	35.31 / 0.976 / 0.059
	Fish	24.11 / 0.842 / 0.187	23.85 / 0.834 / 0.153	22.41 / 0.809 / 0.171	23.27 / 0.822 / 0.151	28.97 / 0.896 / 0.111
	FurScene	24.12 / 0.861 / 0.162	24.77 / 0.868 / 0.113	26.09 / 0.891 / 0.103	25.15 / 0.877 / 0.112	24.47 / 0.864 / 0.119
	Pikachu	17.27 / 0.833 / 0.201	31.06 / 0.962 / 0.055	31.16 / 0.967 / 0.057	30.69 / 0.965 / 0.058	31.65 / 0.967 / 0.056
	Pixiu	19.95 / 0.848 / 0.172	20.70 / 0.841 / 0.143	23.72 / 0.864 / 0.117	23.39 / 0.862 / 0.119	26.20 / 0.881 / 0.107
GS ³ (synthetic)	AnisoMetal	24.32 / 0.923 / 0.072	27.51 / 0.961 / 0.039	27.51 / 0.950 / 0.058	28.00 / 0.957 / 0.047	27.15 / 0.956 / 0.040
	Drums	15.44 / 0.832 / 0.238	20.64 / 0.922 / 0.096	31.52 / 0.976 / 0.029	31.54 / 0.976 / 0.029	30.45 / 0.976 / 0.030
	FurBall	17.61 / 0.890 / 0.169	35.56 / 0.968 / 0.062	35.83 / 0.968 / 0.079	35.00 / 0.965 / 0.086	36.11 / 0.971 / 0.054
	Hotdog	29.40 / 0.953 / 0.060	29.67 / 0.966 / 0.046	33.32 / 0.974 / 0.036	32.94 / 0.973 / 0.037	35.22 / 0.980 / 0.029
	Lego	16.78 / 0.809 / 0.200	20.27 / 0.852 / 0.192	30.91 / 0.956 / 0.050	29.72 / 0.944 / 0.061	30.80 / 0.959 / 0.044
	Translucent	29.29 / 0.961 / 0.052	29.03 / 0.971 / 0.041	32.76 / 0.975 / 0.042	31.99 / 0.975 / 0.038	34.67 / 0.981 / 0.031
SSS-GS (synthetic)	Bunny	37.85 / 0.988 / 0.020	40.43 / 0.991 / 0.016	34.42 / 0.980 / 0.032	38.49 / 0.989 / 0.019	39.86 / 0.990 / 0.016
	Candle	41.88 / 0.994 / 0.015	45.08 / 0.997 / 0.008	38.74 / 0.988 / 0.022	38.74 / 0.989 / 0.020	45.20 / 0.996 / 0.008
	Dragon	35.23 / 0.973 / 0.042	37.79 / 0.982 / 0.026	36.10 / 0.977 / 0.032	37.38 / 0.982 / 0.027	37.38 / 0.981 / 0.026
	Soap	40.28 / 0.991 / 0.013	41.42 / 0.992 / 0.011	38.35 / 0.984 / 0.027	38.36 / 0.988 / 0.022	40.91 / 0.992 / 0.012
	Statue	30.23 / 0.968 / 0.048	38.99 / 0.989 / 0.022	36.43 / 0.985 / 0.025	35.99 / 0.983 / 0.027	39.82 / 0.989 / 0.020

Table S2. **Transfer share and light novelty per scene.** Share of linear radiance routed through L_{tr} by the final model (mean and range over 8 evenly spaced test views), mean learned visibility over covered pixels, and the angle between each official test light and the nearest training light (median / maximum).

Benchmark	Scene	Transfer share	Range	Mean V	Test-light novelty
NRHints (real)	Cat	34.2%	28–44%	0.66	2.3° / 10.5°
	CatSmall	26.8%	21–40%	0.55	1.2° / 3.5°
	CupFabric	19.2%	12–25%	0.76	1.3° / 3.6°
	Fish	28.2%	1–45%	0.88	2.2° / 8.8°
	FurScene	7.2%	5–11%	0.59	2.1° / 6.6°
	Pikachu	12.3%	9–20%	0.55	0.9° / 3.7°
	Pixiu	42.1%	37–45%	0.83	2.1° / 5.1°
GS ³ (synthetic)	AnisoMetal	7.4%	6–11%	0.89	1.1° / 3.9°
	Drums	5.6%	4–7%	0.77	1.1° / 3.9°
	FurBall	16.9%	16–19%	0.84	1.1° / 3.9°
	Hotdog	6.7%	6–8%	0.92	1.1° / 3.9°
	Lego	9.3%	7–14%	0.75	1.1° / 3.9°
	Translucent	16.7%	14–21%	0.89	1.1° / 3.9°
SSS-GS (synthetic)	Bunny	28.1%	13–81%	0.67	0.0° / 11.3°
	Candle	47.6%	28–88%	0.49	0.0° / 11.2°
	Dragon	48.2%	34–79%	0.46	0.0° / 11.2°
	Soap	58.0%	32–98%	0.63	0.0° / 11.3°
	Statue	20.5%	10–57%	0.36	0.0° / 11.0°

and the Translucent scene (about 17%), and highest on the subsurface-scattering scenes (20.5–58.0%). On real captures it varies between 7.2% (FurScene) and 42.1% (Pixiu, a translucent resin figure); there the transfer term also absorbs effects that the local model cannot explain. These shares describe how the learned model allocates radiance, not a physical decomposition.

Light novelty of the benchmarks. The last column of Tab. S2 measures, for every official test view, the angle between its light direction and the closest training light direction, both seen from the scene origin. The median is 1.1° on the GS³ scenes, 0.9–2.3° on the real captures and 0° on SSS-GS, where most test lights coincide with training lights

Table S3. **Development history** of LiSA on the same 18-scene protocol (PSNR / LPIPS). Initial: display-domain loss and pixel receivers throughout. Revised: adds a stronger mask loss, the highlight lobes and gated background supervision. Final: the staged schedule of Sec. 4, which also changes initialization, normals and receivers.

Version	Real (7)	GS ³ (6)	SSS (5)	All (18)
Initial (30k)	28.47 / 0.099	28.35 / 0.062	39.07 / 0.020	31.38 / 0.065
Revised (30k)	29.49 / 0.096	28.64 / 0.059	41.03 / 0.015	32.41 / 0.062
Final (38k)	29.28 / 0.100	32.40 / 0.038	40.63 / 0.016	33.47 / 0.056

and only the camera–light combination is new. The benchmarks therefore measure relighting under densely sampled lights rather than extrapolation, which motivates experiment E1.

Development history. Table S3 lists earlier versions of LiSA evaluated with the same protocol. The initial version used the display-domain loss and pixel receivers throughout; the revised version added a stronger mask loss, the highlight lobes and gated background supervision. The final, staged version is far better on the GS³ scenes but slightly behind the revised version on real and SSS scenes; because the two versions also differ in initialization, normals and receivers, this gap cannot be attributed to the loss domain alone.

Training time. Table S4 lists the wall-clock training time of every run in the benchmark. For LiSA it covers both training processes including initialization and saving; for GS³ and SSD-GS it covers the training function up to the last progress record; for RNG both stages; for OLAT-GS both stages including mesh export, with up to 15 s of polling delay. GPU load and concurrency were not controlled uniformly across these runs, so they only indicate the order of

Table S4. **Wall-clock training time** (minutes) of every run in our benchmark. Runs were not timed under strictly uniform conditions (see text); “—” marks SSD-GS runs reused without a timing log. Table 6 gives exclusive-GPU measurements.

Scene	LiSA	GS ³	SSD-GS	OLAT-GS	RNG
Cat	25.1	53.9	69.2	67.8	115.6
CatSmall	20.4	49.1	56.4	59.1	182.3
CupFabric	19.8	52.6	57.8	69.1	170.2
Fish	19.2	61.3	67.5	53.8	82.7
FurScene	20.0	54.5	60.8	70.1	81.8
Pikachu	20.5	53.0	61.7	74.8	175.9
Pixiu	19.7	50.3	59.6	55.8	80.2
AnisoMetal	28.3	63.2	—	78.3	204.5
Drums	26.2	74.8	136.6	81.6	99.6
FurBall	21.5	41.7	59.8	59.1	89.5
Hotdog	21.4	42.6	78.2	75.8	97.3
Lego	26.6	59.0	161.7	103.9	115.7
Translucent	24.1	54.7	—	71.6	237.5
Bunny	16.9	105.2	—	67.8	40.4
Candle	16.8	41.0	27.6	58.1	36.9
Dragon	16.8	43.3	—	57.8	38.2
Soap	16.6	40.6	27.4	61.8	38.5
Statue	16.7	41.3	28.8	71.3	41.1
Mean	20.9	54.5	68.1	68.8	107.1

magnitude; Tab. 6 gives controlled measurements.

S4. Additional qualitative results

Figure S1 shows the atlas and the radiance components for an opaque scene with strong cast shadows and for a real translucent object. Figure S2 illustrates the transfer ablation of Tab. 2, and Fig. S3 the failure cases discussed in Sec. 5.6.

S5. Experiments to be run

The following experiments are referenced in the main paper with red placeholders. They use the frozen final configuration; nothing is tuned on their results.

E1: held-out light directions. Partition every training set into 30° elevation–azimuth bins of the light direction and hold out whole bins, in a fixed order, until about 10% of the training images are held out (the deterministic split implemented in our data loader). On the development scenes this yields held-out lights 4–11° (median) and up to 16° away from the remaining training lights, compared with 0–2.3° for the official test splits. Train LiSA, its local-residual variant, GS³, SSD-GS and OLAT-GS on the remaining images with their default schedules on AnisoMetal, Translucent, Bunny, Dragon and Pixiu (optionally all 18 scenes), and evaluate on the held-out images with the main protocol. Report PSNR/SSIM/LPIPS per scene and the drop relative to the official test split, and plot per-view error against angular novelty.

E2: calibration-disentangled real-scene evaluation. (a) Train LiSA without the rotation refinement of training cam-

eras on the seven real captures. (b) For all five methods, report a secondary score after aligning each test rendering to its target by the best global 2D translation within 24 px, and optionally after a rotation-only test-pose alignment with the scene frozen. Fixed-calibration numbers remain the primary metric.

E3: per-Gaussian visibility. Replace the atlas shadow test by per-Gaussian visibility splatted as an attribute, as in GS³ and SSD-GS (our renderer provides it as an option), while keeping the atlas for the transfer term; same five scenes and three seeds as Tab. 2.

E4: optimization ablation at the full protocol. On the five ablation scenes with three seeds: (a) display-domain loss in all stages; (b) no warm-up, i.e., 30k joint iterations from the hull initialization; (c) pixel receivers in stage II; (d) no stage III, evaluating the 30k checkpoint with Gaussian receivers.

E5: additional baselines. SSS-GS on the SSS-GS scenes once a stable reproduction under our protocol is available, and BiGS if an all-lights-on image for its initialization can be synthesized from the training data.

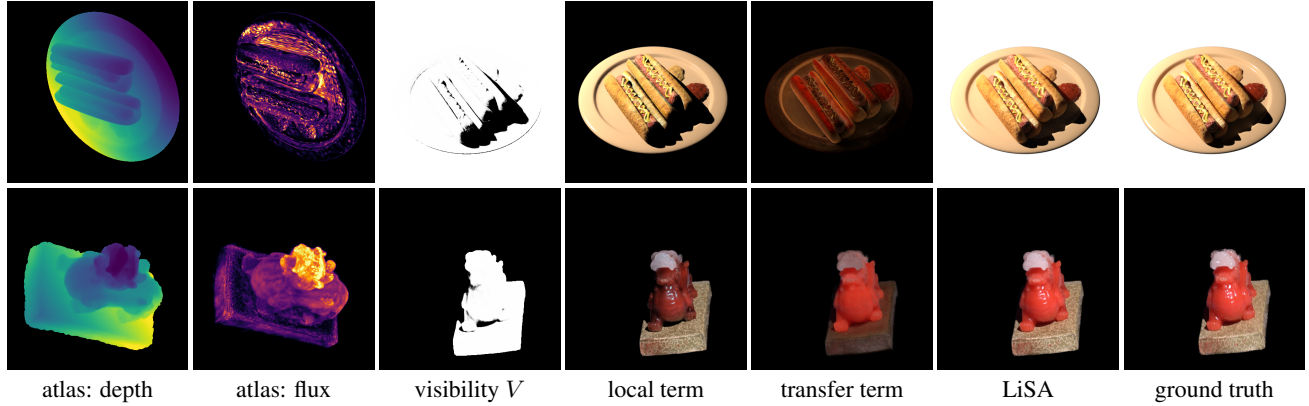


Figure S1. **Atlas and radiance components** of the final models on test views of Hotdog (top, 6.7% of the radiance through the transfer term on average) and of the real Pixiu capture (bottom, a translucent resin figure, 42.1%). On Pixiu the transfer term carries the glow of the resin, while the cast shadow on the box is explained by the visibility.

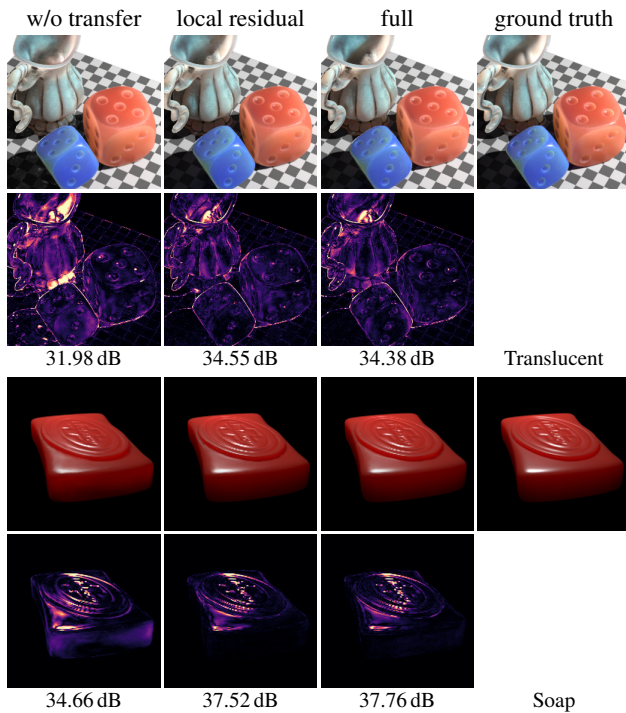


Figure S2. **Transfer ablation** on the first test view of Translucent and Soap (seed 0; PSNR of the full view; error maps show the mean absolute RGB error, saturating at 0.15). Overall brightness is nearly identical across variants, but without the transfer term the errors concentrate on the translucent bodies: the blue die, the base of the vase and the sides of the soap. The local residual and the full model have similar errors, consistent with Tab. 3.

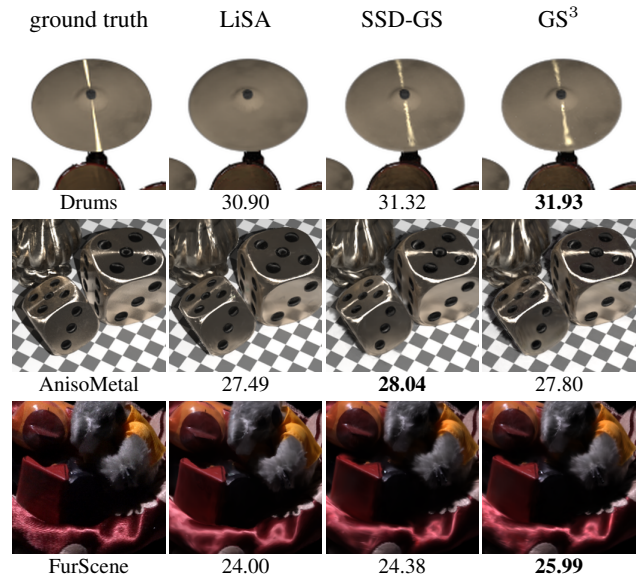


Figure S3. **Failure cases** at the median of the per-view margin (PSNR of the full view). LiSA misses the thin anisotropic streak on the cymbal, renders the brushed-metal dice too uniformly, and loses detail in real fur.